

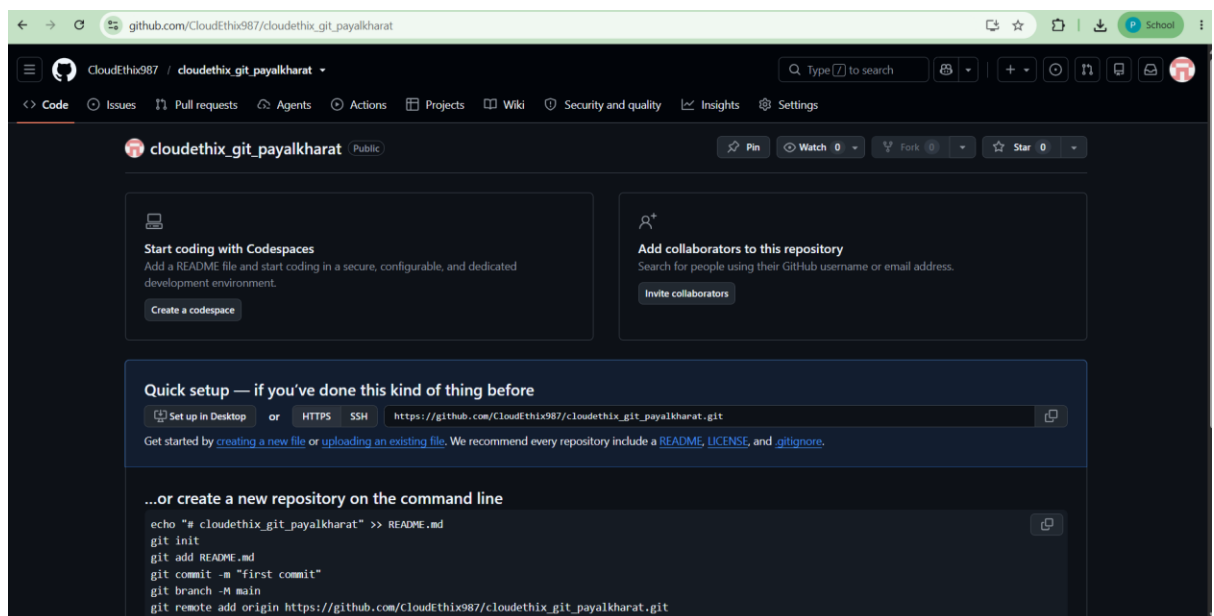
GIT - LAB-02:

Task-1:

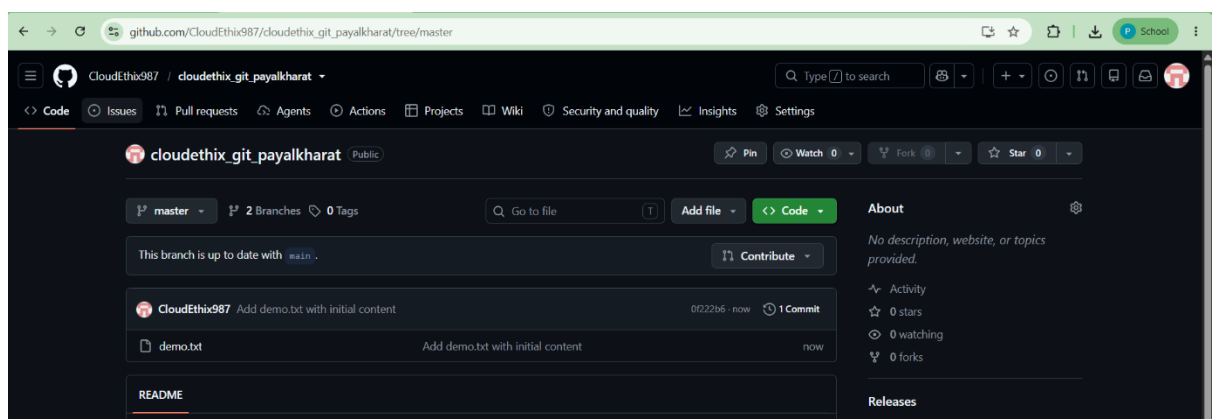
Create a remote "repository" with project name cloudethix_git_yourname. Create one file named demo.txt in the master branch of the remote repository. Once a file is created, clone the repository locally. On the local repository create 3 branches: feature/login and feature/mainline, release. Add login.txt file to feature/login branch and push that branch to remote repository. Create PR and approve the PR and then merge the feature/login branch to the remote master branch.

Implementation:

1. Created remote repo with name cloudethix_git_payalkharat.



2. Added demo.txt file on master branch.



3 . Remote repo is cloned locally.

```
payal_kharat@LAPTOP-V1NV0QFT:~$  
payal_kharat@LAPTOP-V1NV0QFT:~$ git clone https://github.com/CloudEthix987/cloudethix_git_payalkharat.git  
Cloning into 'cloudethix_git_payalkharat'...  
remote: Enumerating objects: 3, done.  
remote: Counting objects: 100% (3/3), done.  
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (3/3), done.  
payal_kharat@LAPTOP-V1NV0QFT:~$ cd cloudethix_git_payal  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payal$ ls  
README.md demo.txt fct.txt git.txt main.txt remote.txt  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payal$ cd ..  
payal_kharat@LAPTOP-V1NV0QFT:~$ cd cloudethix_git_payalkharat  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ ls  
demo.txt
```

4 . On the local repository create 3 branches: feature/login and feature/mainline, release.

```
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch feature/login  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch feature/mainline  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch release  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch  
feature/login  
feature/mainline  
* main  
release
```

5 . Added login.txt file to feature/login branch.

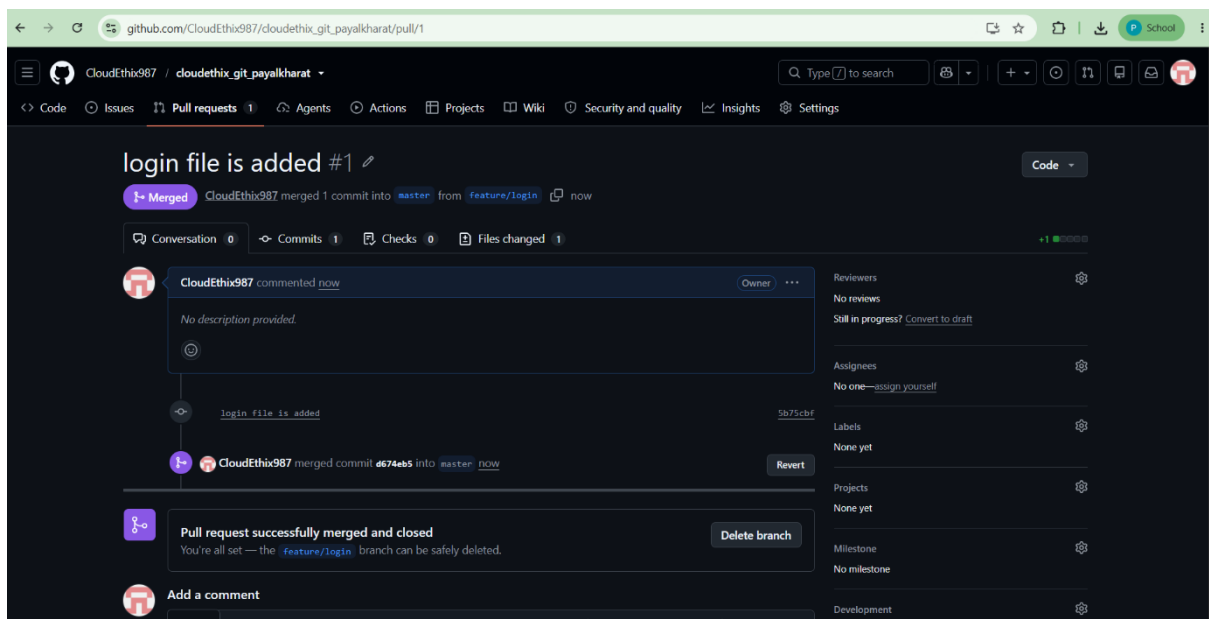
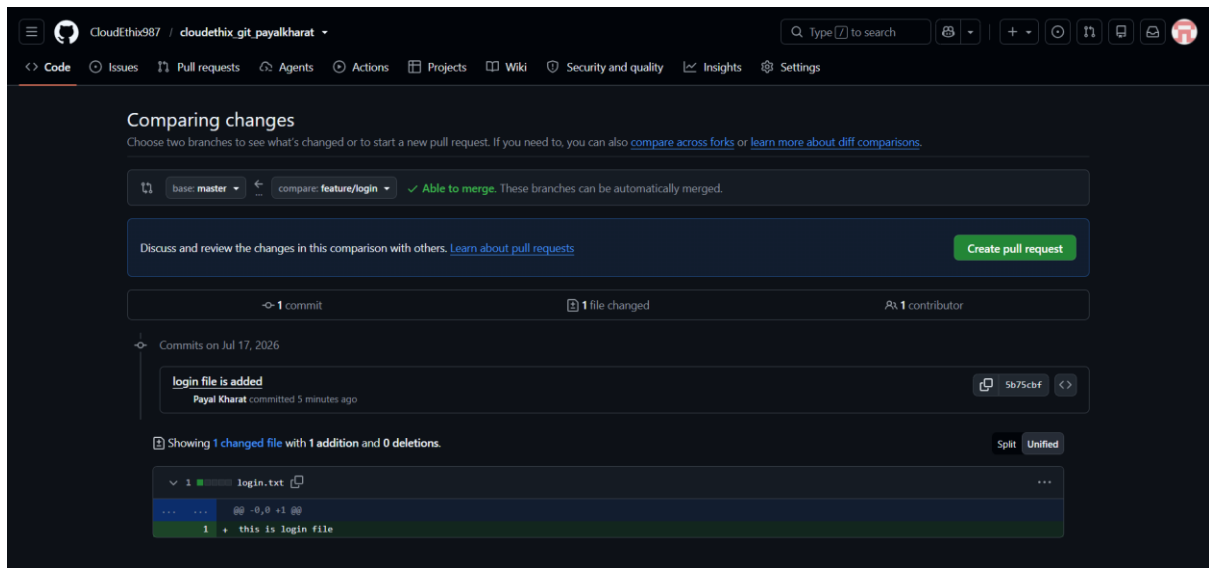
```
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git checkout feature/login  
Switched to branch 'feature/login'  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch  
* feature/login  
feature/mainline  
main  
release  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ echo " this is login file" > login.txt
```

6 . feature/login branch is pushed to remote repo.

```
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ echo " this is login file" > login.txt  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git status  
On branch feature/login  
Untracked files:  
  (use "git add <file>..." to include in what will be committed)  
login.txt  
  
nothing added to commit but untracked files present (use "git add" to track)  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git add login.txt  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git commit -m "login file is added"  
[feature/login 5b75cbf] login file is added  
1 file changed, 1 insertion(+)  
create mode 100644 login.txt
```

```
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git remote add origin https://github.com/CloudEthix987/cloudethix_git_payalkharat.git  
error: remote origin already exists.  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git push -u origin feature/login  
Username for 'https://github.com': CloudEthix987  
Password for 'https://CloudEthix987@github.com':  
Enumerating objects: 4, done.  
Counting objects: 100% (4/4), done.  
Delta compression using up to 12 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 302 bytes | 302.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
remote:  
remote: Create a pull request for 'feature/login' on GitHub by visiting:  
remote: https://github.com/CloudEthix987/cloudethix_git_payalkharat/pull/new/feature/login  
remote:  
To https://github.com/CloudEthix987/cloudethix_git_payalkharat.git  
 * [new branch] feature/login -> feature/login  
branch 'feature/login' set up to track 'origin/feature/login'.  
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ |
```

7 . Create the PR and merge the feature/login branch to the remote master branch.



github.com/CloudEthix987/cloudethix_git_payalkharat/tree/master

CloudEthix987 / cloudethix_git_payalkharat

Search

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Insights

Settings

cloudethix_git_payalkharat

Public

Pin

Watch 0

Fork 0

Star 0

master had recent pushes 41 seconds ago

Compare & pull request

master

3 Branches

0 Tags

Go to file

Add file

Code

This branch is 2 commits ahead of main.

Contribute

CloudEthix987 Merge pull request #1 from CloudEthix987/feature/login

d574eb5 · now

3 Commits

demo.txt

Add demo.txt with initial content

11 minutes ago

login.txt

login file is added

7 minutes ago

README

Add a README

Help people interested in this repository understand your project.

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

CloudEthix987

Task-2:

Add files to the feature/mainline branch to local repo and push that branch to remote. Then add some files to the feature/mainline remote branch from the UI. Then fetch the latest changes on local feature/mainline with Fetch and Pull command.

Implementation:

1 . files are added to the feature/mainline branch to local repo.

```
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch
* feature/login
  feature/mainline
  main
  release
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git checkout feature/mainline
Switched to branch 'feature/mainline'
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch
* feature/login
  feature/mainline
  main
  release
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ touch local_file1.txt
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ touch local_file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git status
On branch feature/mainline
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        local_file1.txt
        local_file2.txt

nothing added to commit but untracked files present (use "git add" to track)
```

2 . Added some files to the feature/mainline remote branch from the UI.

The screenshot shows the GitHub interface for the repository 'cloudethix_git_payalkharat'. The 'feature/mainline' branch is selected. The page displays recent pushes, a commit history table, and repository statistics.

Commit	Author	Files	Time
a1c20a6	Payal Kharat	2 files are added	1 minute ago
		demo.txt	35 minutes ago
		local_file1.txt	1 minute ago
		local_file2.txt	1 minute ago

3 . Fetch the latest changes on local feature/mainline with Fetch and Pull command.

```
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git fetch origin
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 2.66 KiB | 389.00 KiB/s, done.
From https://github.com/CloudEthix987/cloudethix_git_payalkharat
   alc20a6..873944c  feature/mainline -> origin/feature/mainline
   0f222b6..d674eb5  master          -> origin/master
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git pull feature/mainline
fatal: 'feature/mainline' does not appear to be a git repository
fatal: Could not read from remote repository.

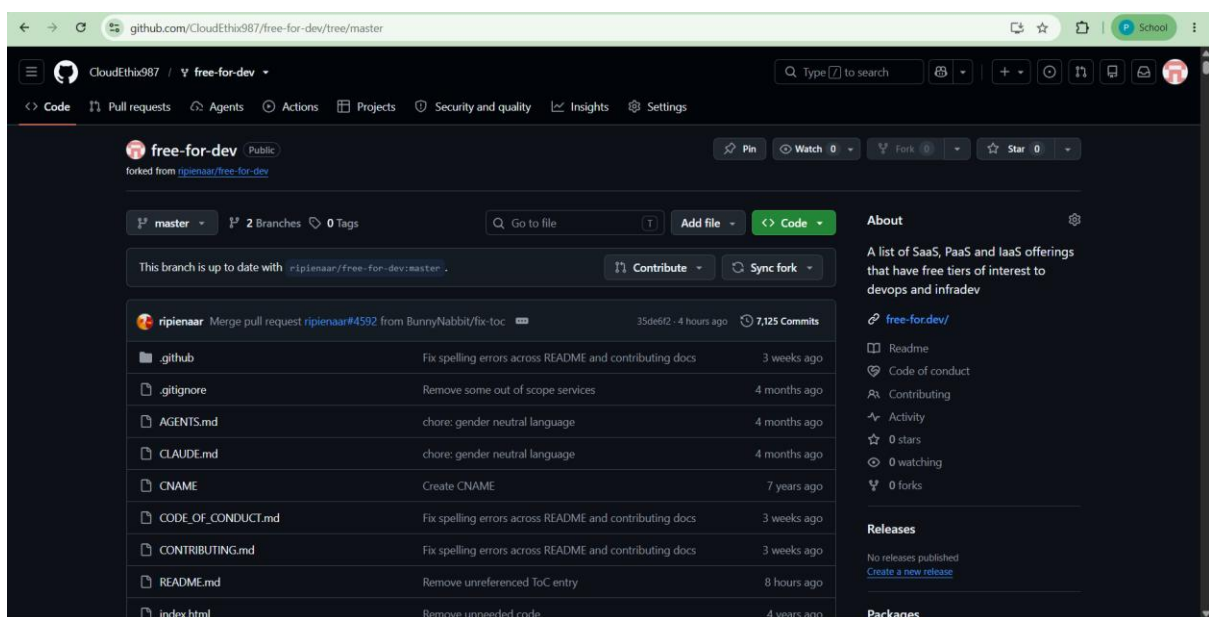
Please make sure you have the correct access rights
and the repository exists.
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git branch
  feature/login
* feature/mainline
  main
  release
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ git pull origin feature/mainline
From https://github.com/CloudEthix987/cloudethix_git_payalkharat
 * branch                feature/mainline -> FETCH_HEAD
Updating alc20a6..873944c
Fast-forward
 remote_file1.txt | 1 +
 remote_file2.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 remote_file1.txt
 create mode 100644 remote_file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ ls
demo.txt  local_file1.txt  local_file2.txt  remote_file1.txt  remote_file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/cloudethix_git_payalkharat$ |
```

Task-3:

Fork below public repository to your account. Then clone the repository locally. Create 2 branches cloud_dev01. Then add cloud_dev02 to that repository locally and hello.txt , moto.txt and world.txt to one branch and add another 3 files to another branch. Then push cloud_dev02 branch to the Forked repository on the remote. [GitHub - ripienaar/free-for-dev: A list of SaaS, PaaS and IaaS offerings that have free tiers of int...](#)

Implementation:

1 . Forked the given repo to my account.



2 . Repo cloned to locally.

```
payal_kharat@LAPTOP-V1NV0QFT:~$ git clone https://github.com/CloudEthix987/free-for-dev.git
Cloning into 'free-for-dev'...
remote: Enumerating objects: 16781, done.
remote: Counting objects: 100% (38/38), done.
remote: Compressing objects: 100% (27/27), done.
remote: Total 16781 (delta 31), reused 11 (delta 11), pack-reused 16743 (from 4)
Receiving objects: 100% (16781/16781), 10.06 MiB | 7.29 MiB/s, done.
Resolving deltas: 100% (8922/8922), done.
payal_kharat@LAPTOP-V1NV0QFT:~$ ls
CloudEthixDevops  Git_Demo_Project  Student_Website  cloudeithix_git_payal  free-for-dev
Devops_Project   Pandharpur_project cloudeithix_devops_payalkharat cloudeithix_git_payalkharat git_branch_demo
payal_kharat@LAPTOP-V1NV0QFT:~$ cd free-for-dev/
```

3 . Create 2 new branch.

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch
* master
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch cloud_dev01
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch cloud_dev02
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch
cloud_dev01
cloud_dev02
* master
```

4 . On cloud-dev01 branch 3 files are added.

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git checkout cloud_dev02
Switched to branch 'cloud_dev02'
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch
  cloud_dev01
* cloud_dev02
  master
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ touch hello.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CLAUDE.md  CNAME      CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  hello.txt  index.html  logo.webp
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ touch moto.txt world.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CLAUDE.md  CNAME      CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  hello.txt  index.html  logo.webp  moto.txt  world.txt
```

5 . On cloud_dev02 branch another 3 files are added.

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git checkout cloud_dev01
Switched to branch 'cloud_dev01'
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ touch file1.txt file2.txt file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CNAME      CONTRIBUTING.md  file1.txt  file3.txt  index.html  moto.txt
CLAUDE.md  CODE_OF_CONDUCT.md  README.md      file2.txt  hello.txt  logo.webp   world.txt
```

6 .cloud_dev02 branch is pushed to the Forked repository on the remote.

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git add .
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git status
On branch cloud_dev02
Your branch is up to date with 'origin/cloud_dev02'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   file1.txt
    new file:   file2.txt
    new file:   file3.txt
    new file:   hello.txt
    new file:   moto.txt
    new file:   world.txt

payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch -m "new files are added"
fatal: 'new files are added' is not a valid branch name
hint: See 'git help check-ref-format'
hint: Disable this message with "git config set advice.refSyntax false"
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git commit -m "new files are added"
[cloud_dev02 99229204] new files are added
 6 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1.txt
 create mode 100644 file2.txt
 create mode 100644 file3.txt
 create mode 100644 hello.txt
 create mode 100644 moto.txt
 create mode 100644 world.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git push -u origin cloud_dev02
Username for 'https://github.com': CloudEthix987
Password for 'https://CloudEthix987@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 323 bytes | 323.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/CloudEthix987/free-for-dev.git
   35de6f23..99229204  cloud_dev02 -> cloud_dev02
branch 'cloud_dev02' set up to track 'origin/cloud_dev02'.
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$
```


github.com/CloudEthix987/free-for-dev/tree/cloud_dev02

Payal Kharat

new files are added

9922920 · 2 minutes ago

7,126 Commits

.github	Fix spelling errors across README and contributing docs	3 weeks ago
.gitignore	Remove some out of scope services	4 months ago
AGENTS.md	chore: gender neutral language	4 months ago
CLAUDE.md	chore: gender neutral language	4 months ago
CNAME	Create CNAME	7 years ago
CODE_OF_CONDUCT.md	Fix spelling errors across README and contributing docs	3 weeks ago
CONTRIBUTING.md	Fix spelling errors across README and contributing docs	3 weeks ago
README.md	Remove unreferenced ToC entry	8 hours ago
file1.txt	new files are added	2 minutes ago
file2.txt	new files are added	2 minutes ago
file3.txt	new files are added	2 minutes ago
hello.txt	new files are added	2 minutes ago
index.html	Remove unneeded code	4 years ago
logo.webp	Add files via upload	5 years ago
moto.txt	new files are added	2 minutes ago
world.txt	new files are added	2 minutes ago

Code of conduct

Contributing

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors

No contributors

Languages

HTML 100.0%

Suggested workflows

Based on your tech stack

Jekyll using

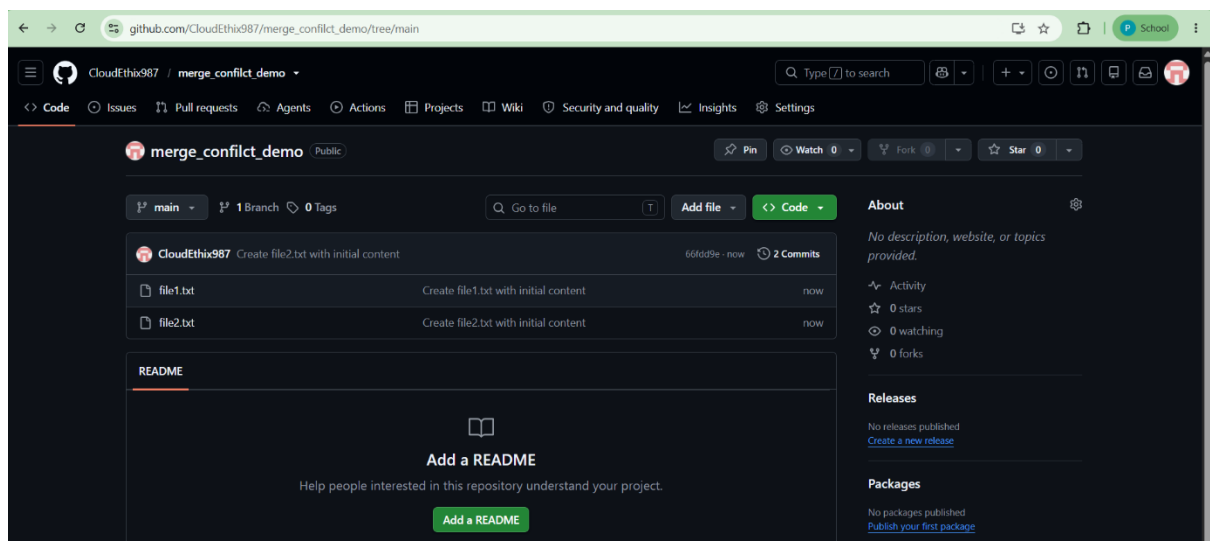
Configure

Task-4:

Create a remote repository merge_confilct_demo. Add some files in the remote master branch. Clone the repository locally. Create 2 branches locally named demo/merge and demo/conflicts. Add a merge.txt file in both branches with some data like "Hello CloudEthiX...." And "Fello CloudEthiX...." respectively and commit. Then merge demo/conflicts branch to demo/merge branch and solve the merge conflicts.

Implementation:

1 . Created a remote repository merge_confilct_demo and Add files in the remote main branch.



2 . Repo cloned to local and 2 branches are created.

```
payal_kharat@LAPTOP-V1NV0QFT:~$ git clone https://github.com/CloudEthix987/merge_confilct_demo.git
Cloning into 'merge_confilct_demo'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
payal_kharat@LAPTOP-V1NV0QFT:~$ cd merge_confilct_demo/
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git branch
* main
```

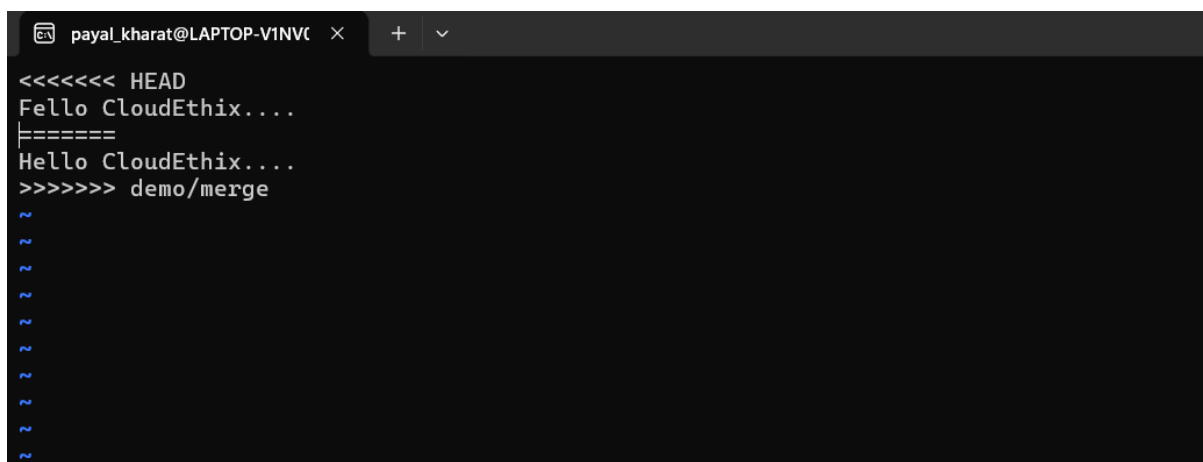
```
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git branch
* main
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git branch demo/merge
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git branch demo/conflicts
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git branch
demo/conflicts
demo/merge
* main
```

3. Added merge.txt file in both branches with some data and commit.

```
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git checkout demo/conflicts
Switched to branch 'demo/conflicts'
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ echo "Hello CloudEthix ...." > merge.txt
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ ls
file1.txt file2.txt merge.txt
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git checkout demo/m
error: pathspec 'demo/m' did not match any file(s) known to git
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ echo "Fello CloudEthix ...." > merge.txt
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ ls
file1.txt file2.txt merge.txt
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git add merge.txt
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git commit -m "merge.txt file added"
[demo/conflicts 05486fb] merge.txt file added
1 file changed, 1 insertion(+)
create mode 100644 merge.txt
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git checkout demo/c
error: pathspec 'demo/c' did not match any file(s) known to git
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git checkout demo/conflicts
Already on 'demo/conflicts'
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git checkout demo/merge
Switched to branch 'demo/merge'
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git add .
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git commit -m "merge file is added"
On branch demo/merge
nothing to commit, working tree clean
```

4 . After merging this two branches conflict is arises.

```
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git merge demo/merge
Auto-merging merge.txt
CONFLICT (add/add): Merge conflict in merge.txt
Automatic merge failed; fix conflicts and then commit the result.
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ vim merge.txt
```



```
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ vim merge.txt
<<<<<< HEAD
Fello CloudEthix....
=====
Hello CloudEthix....
>>>>>> demo/merge
```

5. After solving conflict data is merged.

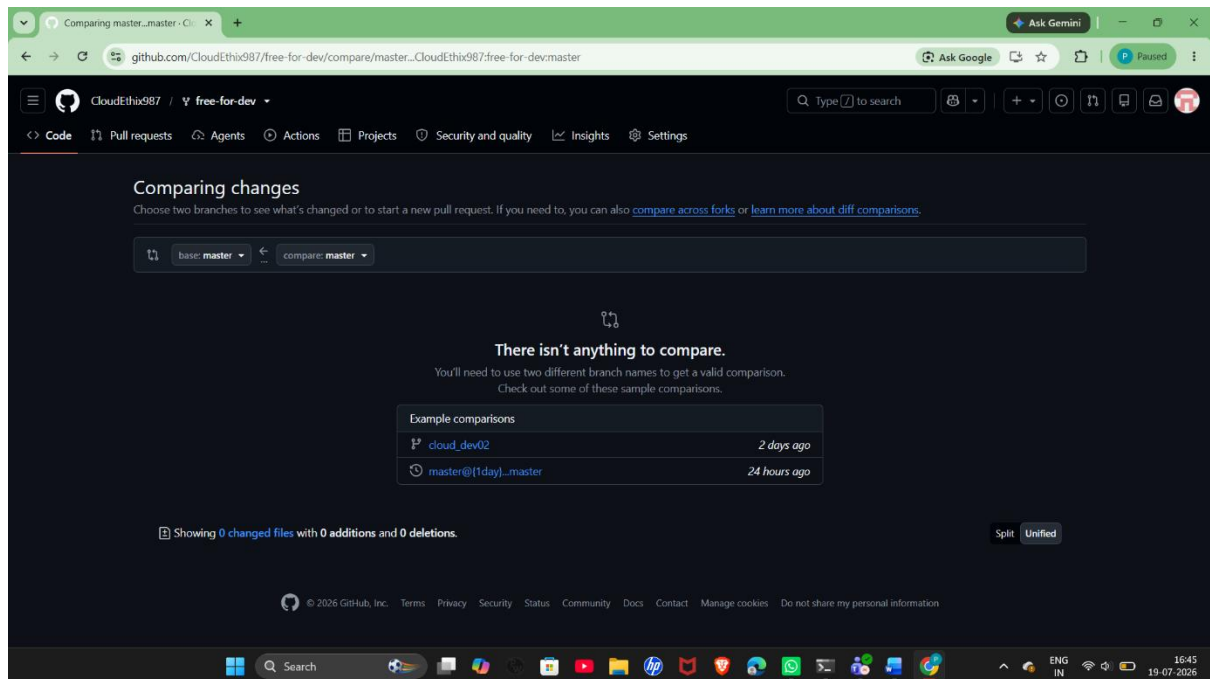
```
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git add .
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ git commit -m "conflict solved"
[demo/conflicts 2e5daaf] conflict solved
payal_kharat@LAPTOP-V1NV0QFT:~/merge_confilct_demo$ cat merge.txt
Fello CloudEthix....
Hello CloudEthix....
```

Task – 5:

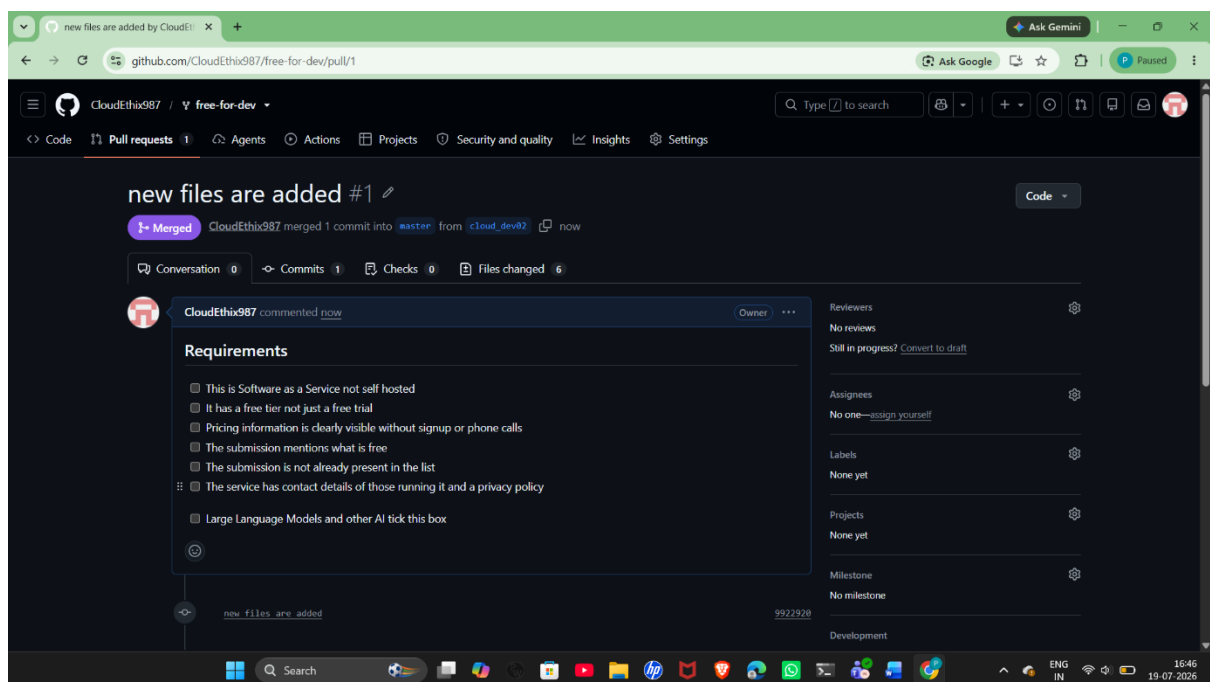
Create PR to merge the cloud_dev02 branch to Forked repository. Approve the PR and merge cloud_dev02 branch to master branch remotely.

Implementation:

1 . One pull request is created .



2 . Master branch is merged into cloud_dev02 branch.



CloudEthix987/free-for-dev: A

Ask Gemini

github.com/CloudEthix987/free-for-dev

Ask Google

Paused

CloudEthix987 / free-for-dev

Type to search

Code

Pull requests

Agents

Actions

Projects

Security and quality

Insights

Settings

free-for-dev

Public

forked from rpienaar/free-for-dev

Pin

Watch 0

Fork 0

Star 0

master

2 Branches

0 Tags

Go to file

Add file

Code

This branch is 2 commits ahead of and 18 commits behind rpienaar/free-for-dev:master.

Contribute

Sync fork

CloudEthix987

Merge pull request #1 from CloudEthix987/cloud_dev02

128c35f · 1 minute ago

7,127 Commits

.github

Fix spelling errors across README and contributing docs

3 weeks ago

.gitignore

Remove some out of scope services

4 months ago

AGENTS.md

chore: gender neutral language

4 months ago

CLAUDE.md

chore: gender neutral language

4 months ago

CNAME

Create CNAME

7 years ago

CODE_OF_CONDUCT.md

Fix spelling errors across README and contributing docs

3 weeks ago

CONTRIBUTING.md

Fix spelling errors across README and contributing docs

3 weeks ago

README.md

Remove unreferenced ToC entry

2 days ago

file1.txt

new files are added

2 days ago

About

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Task – 6:

Fetch latest changes from master branch of remote forked repository to local master branch. Then merge forked remote master branch with local master branch. Check If all the latest files are there. Then checkout to local cloud_dev03 branch locally. Then Pull latest changes from the remote master branch.

Implementation:

1 . Changes are fetched from remote forked repo to local master branch .

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch
cloud_dev01
* cloud_dev02
master
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch
cloud_dev01
cloud_dev02
* master
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git fetch origin
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 912 bytes | 912.00 KiB/s, done.
From https://github.com/CloudEthix987/free-for-dev
35de6f23..128c95f4 master -> origin/master
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git status
On branch master
Your branch is behind 'origin/master' by 2 commits, and can be fast-forwarded.
(use "git pull" to update your local branch)

nothing to commit, working tree clean
```

2 . Forked master branch is merged with local master branch.

```
nothing to commit, working tree clean
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git merge origin/master
Updating 35de6f23..128c95f4
Fast-forward
 file1.txt | 0
 file2.txt | 0
 file3.txt | 0
 hello.txt | 0
 moto.txt  | 0
 world.txt | 0
6 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file1.txt
create mode 100644 file2.txt
create mode 100644 file3.txt
create mode 100644 hello.txt
create mode 100644 moto.txt
create mode 100644 world.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CNAME  CONTRIBUTING.md  file1.txt  file3.txt  index.html  moto.txt
CLAUDE.md  CODE_OF_CONDUCT.md  README.md  file2.txt  hello.txt  Logo.webp  world.txt
```

3 . Checkout to cloud_dev03 branch pull the changes from remote master branch.

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git branch cloud_dev03
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git checkout cloud_dev03
Switched to branch 'cloud_dev03'
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git pull origin cloud_dev03
fatal: couldn't find remote ref cloud_dev03
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git pull origin master
From https://github.com/CloudEthix987/free-for-dev
 * branch            master       -> FETCH_HEAD
Already up to date.
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CNAME                CONTRIBUTING.md  file1.txt  file3.txt  index.html  moto.txt
CLAUDE.md  CODE_OF_CONDUCT.md  README.md       file2.txt  hello.txt  logo.webp   world.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ |
```

Task – 7:

Checkout to the cloud_dev01 branch locally and rebase the remote master branch of the forked repository.

Implementation:

1 . remote master branch is rebase on cloud_dev01 branch.

```
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git checkout cloud_dev01
Switched to branch 'cloud_dev01'
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CLAUDE.md  CNAME  CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  index.html  logo.webp
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git fetch origin
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ git rebase origin/master
Successfully rebased and updated refs/heads/cloud_dev01.
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ ls
AGENTS.md  CNAME  CONTRIBUTING.md  file1.txt  file3.txt  index.html  moto.txt
CLAUDE.md  CODE_OF_CONDUCT.md  README.md  file2.txt  hello.txt  logo.webp  world.txt
payal_kharat@LAPTOP-V1NV0QFT:~/free-for-dev$ |
```


Task – 8:

Create one repository locally. Commit any 3 files to the local repository. And the revert back the commits.

Implementation:

1 . Local repo is created.

```
payal_kharat@LAPTOP-V1NV0QFT:~$ mkdir git-revert-demo
payal_kharat@LAPTOP-V1NV0QFT:~$ cd git
git-revert-demo/ git_branch_demo/
payal_kharat@LAPTOP-V1NV0QFT:~$ cd git-revert-demo/
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/payal_kharat/git-revert-demo/.git/
```

2 . Files are added to local repo and commit it.

```
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ touch file1.txt file2.txt file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ ls
file1.txt file2.txt file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ echo "Hello File 1" > file1.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ echo "Hello File 2" > file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ echo "Hello File 3" > file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ ls
file1.txt file2.txt file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git add .
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file1.txt
        new file:   file2.txt
        new file:   file3.txt

payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git commit -m "Added three files"
[master (root-commit) 3b9ef8a] Added three files
 3 files changed, 3 insertions(+)
 create mode 100644 file1.txt
 create mode 100644 file2.txt
 create mode 100644 file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git status
On branch master
nothing to commit, working tree clean
```

3 . And Revert back that commit.

```
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git log --oneline
b268119 (HEAD -> master) Added three files
d0d6a37 Revert "Added three files"
3b9ef8a Added three files
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git revert HEAD
[master 7598e8a] Revert "Added three files"
 3 files changed, 3 deletions(-)
 delete mode 100644 file1.txt
 delete mode 100644 file2.txt
 delete mode 100644 file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ git status
On branch master
nothing to commit, working tree clean
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ ls
payal_kharat@LAPTOP-V1NV0QFT:~/git-revert-demo$ |
```

Task – 9:

Create one local repository. commit some files to the local master branch. Checkout to master_02 branch. Create 2 more files. Add those to the staging area. Create stash area and those files. Then list and delete the stash.

Implementation:

1 . One repo is created with name stash_demo.

```
payal_kharat@LAPTOP-V1NV0QFT:~$ mkdir stash-demo
payal_kharat@LAPTOP-V1NV0QFT:~$ cd stash-demo/
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/payal_kharat/stash-demo/.git/
```

2 . Commit some file to local master branch.

```
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ touch file1.txt file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ echo "Hello" > file1.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ echo "Git" > file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git add .
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git commit -m "Initial commit on master"
[master (root-commit) 59cbd03] Initial commit on master
 2 files changed, 2 insertions(+)
 create mode 100644 file1.txt
 create mode 100644 file2.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git checkout -b master_02
```

3 . Checkout to master_02 branch. Create 2 more files.

```
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git checkout -b master_02
Switched to a new branch 'master_02'
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ touch file3.txt file4.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ echo "File 3" > file3.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ echo "File 4" > file4.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git add file3.txt file4.txt
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git status
On branch master_02
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   file3.txt
    new file:   file4.txt
```

4 . Stash area is created , listed and deleted.

```
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git stash
Saved working directory and index state WIP on master_02: 59cbd03 Initial commit on master
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git stash list
stash@{0}: WIP on master_02: 59cbd03 Initial commit on master
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ git stash drop stash@{0}
Dropped stash@{0} (dba8a69391826dd0f911bea1a8aeb687280fe2a7)
payal_kharat@LAPTOP-V1NV0QFT:~/stash-demo$ |
```

Task – 10:

Create a local repository, initialise it and create 3 branches: develop, hotfix and release. Checkout to hotfix branch and commit cloudehix.txt and once committed amend the commit message.

Implementation:

1 . One local repo is created.

```
payal_kharat@LAPTOP-V1NV0QFT:~$ mkdir branch_demo
payal_kharat@LAPTOP-V1NV0QFT:~$ cd branch_demo/
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/payal_kharat/branch_demo/.git/
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ touch file1.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git add .
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git commit -m "one file is added"
[master (root-commit) 400aaf7] one file is added
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git branch
* master
```

2 . Three branches are created.

```
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git branch
* master
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git branch develop
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git branch hotfix
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git branch release
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git branch
  develop
  hotfix
* master
  release
```

3 . Checkout to hotfix branch and commit cloudehith.txt and once committed amend the commit message.

```
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git checkout hotfix
Switched to branch 'hotfix'
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ echo "Cloudehith Assignment" > cloudehith.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ ls
cloudehith.txt  file1.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git add cloudehith.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git commit -m "Added cloudehith file"
[hotfix bb4c352] Added cloudehith file
 1 file changed, 1 insertion(+)
 create mode 100644 cloudehith.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git commit --amend -m "Added cloudehith.txt to hotfix branch"
[hotfix e3695ef] Added cloudehith.txt to hotfix branch
 Date: Sun Jul 19 12:16:49 2026 +0000
 1 file changed, 1 insertion(+)
 create mode 100644 cloudehith.txt
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$ git log --oneline
e3695ef (HEAD -> hotfix) Added cloudehith.txt to hotfix branch
400aaf7 (release, master, develop) one file is added
payal_kharat@LAPTOP-V1NV0QFT:~/branch_demo$
```